

MTH6113 Assessment 1 Report

Introduction

This report analyses a three-stock portfolio consisting of Meta Platforms (META), K Wave Media, and L'Oréal. Using one year of daily closing price data, daily log returns were calculated and used to estimate the average return, variance, standard deviation, covariance, and correlation of the three assets. These statistics were then applied to portfolio analysis to examine diversification benefits, identify inefficient and efficient portfolios, and determine the minimum variance portfolio.

A. Analysis of the Stocks

A.1 Brief Overview of Stocks and Their Industries

Meta Platforms, Inc. (NASDAQ: META)

Meta is one of the world's largest technology companies and owns major social media platforms including Facebook, Instagram, WhatsApp, Messenger, and Threads. The company earns most of its revenue from digital advertising, while also investing heavily in Reality Labs products such as VR headsets and smart glasses. Meta operates in the global digital advertising industry and benefits from its huge user base and AI-driven advertising tools. It reported strong financial performance in 2025, but faces risks from high AI investment costs, regulatory pressure, and uncertainty over future returns. Compared to the other stocks, Meta represents a high growth option, offering more upside than the defensive stability of L'Oréal, while maintaining significantly more institutional backing and scale than the volatility of K Wave Media.

K Wave Media

K Wave Media is a small-cap entertainment company focused on Korean Wave content such as K-pop, K-dramas, and films. It operates in the streaming and digital entertainment industry, which has grown due to the rising global demand for Korean culture. Strategically, K Wave Media has also attempted to integrate itself through acquisitions of entertainment and technology firms; integration of AI and digital media capabilities; even a bitcoin treasury strategy. Compared with Meta and L'Oréal, it is much smaller, less profitable, and significantly more volatile. Its performance is strongly affected by content releases, licensing agreements, and trading liquidity.

L'Oréal (EPA: OR)

L'Oréal is the world's largest beauty and cosmetics company, with brands across luxury, consumer, professional, and dermatological products. It operates in the global beauty industry, which is resilient even during economic downturns. Its growth has been supported by premium beauty demand and expansion in emerging markets. Among the three stocks,

L'Oréal is the most defensive and stable, although it also introduces currency risk for non-euro investors

A.2 Interpretation of the Descriptive Statistics

We can see clear differences in the risk and return characteristics of the three stocks based on the Excel data.

	Mean	Variance
K Wave	-0.00886761	0.02612596
Meta	0.00021005	0.00058399
L'Oreal	-0.00001348	0.00065496

Covariance Matrix

	K Wave	Meta	L'Oreal
K wave	1	0.00021524	0.00031144
Meta	0.00021524	1	0.00010409
L'Oreal	0.00031144	0.00010409	1

K wave Media

- K Wave Media has an expected daily return of approximately **-0.0089**, a variance of **0.0261**, and a standard deviation of **0.1616**.
- Indicating that K Wave Media was a lot more volatile over the same period and returns which were on average negative.
- Overall, it delivered the worst performance while also exposing investors to the greatest uncertainty.

Meta

- Meta has an expected daily return of approximately **0.0002**, a variance of **~0.0006**, and a standard deviation of **~0.0242**.
- Indicating that Meta produced the highest average daily return over the period while maintaining low volatility compared with the other stocks.
- Overall, it delivered the strongest performance in the sample and offered a more favorable risk-return profile than K Wave Media and L'Oréal.

L'oreal

- L'oreal has an expected daily return of approximately **-0.00001348**, a variance of **~0.0007**, and a standard deviation of **~0.0256**.
- Indicating that L'Oréal was fairly stable over the same period, overall lower volatility than K Wave and whilst the average return was negative it was closer to zero

- Overall, it delivered a much more defensive performance than K Wave Media, but it was weaker than Meta in terms of average return.

The covariance values are all positive but small, implying that the stocks move together weakly. This is significant as this low covariance helps reduce portfolio risk through diversification. Most offered through the covariance of Meta and L'oreal (whose covariance is the lowest).

B. Analysis of the Portfolio

B.1 Reasons for Investing in a Portfolio Instead of One Stock Only

Investing in a portfolio reduces unsystematic risk because the variance of a portfolio depends not only on the variances of individual stocks, but also on their covariances. Mathematically, portfolio variance is

$$\sigma_p^2 = \sum w_i^2 \sigma_i^2 + \sum \sum w_i w_j \text{COV}(R_i, R_j)$$

So, risk can be lowered when assets are not perfectly positively related.

A portfolio can produce a more efficient risk-return trade-off than a single stock. The expected return of a portfolio is the weighted average of the individual expected returns,

$$E(R_p) = \sum w_i E(R_i)$$

but its risk does not rise in the same linear way because diversification can reduce variance. This means an investor may achieve a given return with less risk than by holding one stock alone.

Portfolio investing gives access to the efficient frontier, where portfolios offer the highest expected return for each level of risk. A single stock is just one point in risk-return space, but a portfolio creates many combinations of weights. Mathematically, changing

$$w_1, w_2, w_3$$

allows investors to choose portfolios that dominate individual stocks in mean-variance terms.

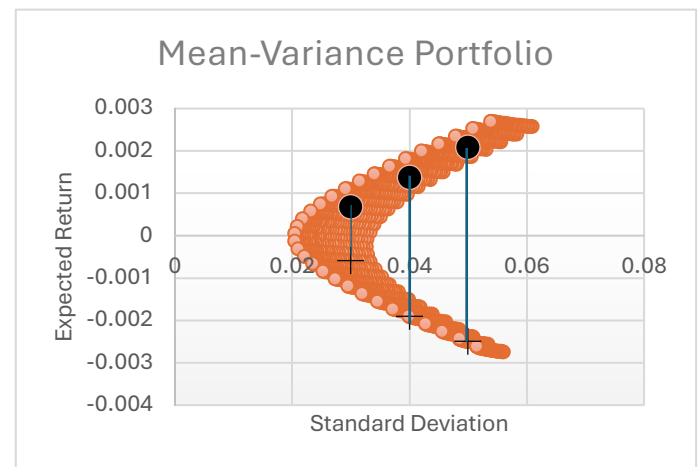
B.2 Inefficient Portfolios and Efficient Alternatives

Efficient portfolios lie above the MVP on the Mean Variance Portfolio, offering higher expected return for the same or higher standard deviation (which can be seen labelled as dots). The three inefficient portfolios (labelled as crosses) sit below the MVP on the frontier, carrying higher risk than the MVP but offer negative expected returns (expected losses). No rational mean-variance investor would choose these portfolios, because for the same standard deviation there exists a portfolio with a higher expected return.

Identifying Portfolios

Three comparisons:

- **Portfolio A_{Cross}** is inefficient because at a standard deviation (SD) of approximately 0.030, it offers an expected return of -0.000580. By moving vertically to the efficient frontier, we find **Portfolio A_{Circle}**, which has the same risk (SD) at 0.030 but provides a much higher expected return of 0.000690
- **Portfolio B_{Cross}** is inefficient because another portfolio with similar risk (SD) provides a significantly better return. Portfolio B has a standard deviation of approximately 0.040 and a return of -0.001899, whereas **Portfolio B_{Circle}** on the efficient frontier provides a return of 0.001376 for that same risk (SD)
- **Portfolio C_{Cross}** is inefficient for the same reason. At a considerable risk (SD) level of approximately 0.050, Portfolio C returns -0.002488, whereas **Portfolio C_{Circle}** expected return is 0.002085 for this risk level



How we found these efficient portfolios

To find these efficient portfolios, I followed a three-step process:

We first identified the Minimum Variance Portfolio with a standard deviation of 0.0204. This point represents the turning point with the lowest standard deviation.

We then split the dataset into two groups: those with returns higher than the Minimum Variance Portfolio (the Efficient Frontier set) and those with lower returns (the Inefficient set).

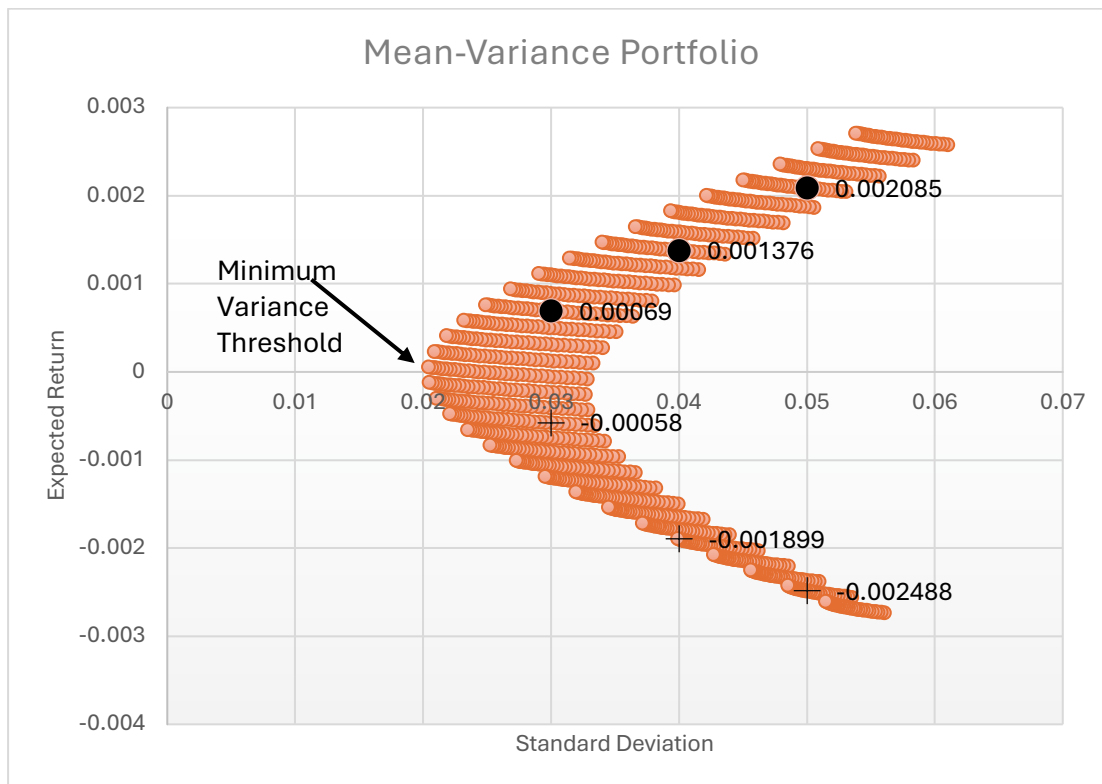
Finally, for a chosen inefficient portfolio (A, B, or C), we searched the efficient set for the portfolio with the matching Standard Deviation. Mathematically, this corresponds to solving for maximum expected returns subject to the standard deviation at various fixed rates.

Diagram Identification

In the scatter plot provided:

- **The + (cross)** represents the inefficient portfolios (located on the lower curve).
- **The ●** represents the efficient portfolios located on the upper boundary.
- **The Lines** represent the vertical move an investor makes to optimise their position, increasing return without increasing risk.

B.3 Minimum Variance Portfolio from the Diagram



Based on our portfolio, the **Minimum Variance Portfolio** is the point that represents the lowest possible risk (standard deviation) achievable with this combination of stocks.

1. Identifying the MVP on the Diagram

The Minimum Variance Portfolio is the leftmost point of the entire portfolio. At this point, the curvature of the boundary changes from the inefficient (lower) half to the efficient (upper) frontier.

- Approximate Risk (Standard Deviation): 0.0204
- Approximate Expected Return: 0.000054

2. Approximate Weights for Each Stock

The analysis of the weights grid indicates that to minimise risk, the portfolio avoids the most volatile asset (K Wave Media) entirely and balances the two more stable assets (Meta and L'Oreal).

Approximate weights for the MVP are:

- Weight 1 (Kenekchukwu / K Wave Media): 0%
- Weight 2 (Morgan / Meta): 30%
- Weight 3 (Lisa / L'Oreal): 70%

This allocation makes financial sense because:

- K Wave Media has a significantly higher individual standard deviation (approximately 0.161) compared to the others. Even lesser amounts of it would drastically increase the total portfolio risk.
- Meta and L'Oreal have much lower individual risks (approximately 0.024 and 0.025 respectively).

The 70% to 30% split between L'Oreal and Meta is the specific mathematical balance that cancels out the most diversifiable risk based on their covariance, resulting in a portfolio risk (0.0204) that is lower than either stock on its own.

B.4 Solving for the minimum variance portfolio analytically

$$\begin{aligned}
 L(\lambda, w) &= \sigma_p^2 + \lambda(1 - w_1 - w_2 - w_3) \\
 &= w_1^2 \text{Var}_1 + w_2^2 \text{Var}_2 + w_3^2 \text{Var}_3 + 2w_1w_2 \text{cov}_{12} + 2w_1w_3 \text{cov}_{13} + 2w_2w_3 \text{cov}_{23} \\
 &\quad + \lambda(1 - w_1 - w_2 - w_3) \\
 \frac{\partial L}{\partial w_1} &= 2w_1 \text{Var}_1 + 2w_2 \text{cov}_{12} + 2w_3 \text{cov}_{13} - \lambda = 0 \\
 &\Rightarrow 2w_1 \text{Var}_1 + 2w_2 \text{cov}_{12} + 2w_3 \text{cov}_{13} = \lambda \\
 \frac{\partial L}{\partial w_2} &= 2w_2 \text{Var}_2 + 2w_1 \text{cov}_{12} + 2w_3 \text{cov}_{23} - \lambda = 0 \\
 &\Rightarrow 2w_2 \text{Var}_2 + 2w_1 \text{cov}_{12} + 2w_3 \text{cov}_{23} = \lambda \\
 \frac{\partial L}{\partial w_3} &= 2w_3 \text{Var}_3 + 2w_1 \text{cov}_{13} + 2w_2 \text{cov}_{23} - \lambda = 0 \\
 &\Rightarrow 2w_3 \text{Var}_3 + 2w_1 \text{cov}_{13} + 2w_2 \text{cov}_{23} = \lambda \\
 \frac{\partial L}{\partial \lambda} &= 1 - w_1 - w_2 - w_3 = 0 \\
 &\Rightarrow w_1 + w_2 + w_3 = 1
 \end{aligned}$$

We know $\Sigma = \begin{matrix} & \begin{matrix} K \\ M \\ L \end{matrix} \end{matrix} \begin{pmatrix} 0.0 & 2.6 & 1.2 & 6.0 & 0.0 & 0.0 & 0.2 & 1.5 & 0.0 & 0.0 & 0.3 & 1.1 \\ 0.0 & 0.0 & 0.2 & 1.5 & 0.0 & 0.0 & 0.5 & 8.4 & 0.0 & 0.0 & 0.1 & 0.4 \\ 0.0 & 0.0 & 0.3 & 1.1 & 0.0 & 0.0 & 0.1 & 0.4 & 0.0 & 0.0 & 0.6 & 5.5 \end{pmatrix}$

$$\begin{aligned}
 &\Rightarrow 2w_1 \text{Var}_1 + 2w_2 \text{cov}_{12} + 2w_3 \text{cov}_{13} = \lambda \\
 &\Rightarrow 2w_2 \text{Var}_2 + 2w_1 \text{cov}_{12} + 2w_3 \text{cov}_{23} = \lambda \\
 &\Rightarrow 2w_3 \text{Var}_3 + 2w_1 \text{cov}_{13} + 2w_2 \text{cov}_{23} = \lambda \\
 &\Rightarrow w_1 + w_2 + w_3 = 1
 \end{aligned}$$

$$\begin{aligned}
 2w_1(0.0 & 2.6 & 1.2 & 6.0) + 2w_2(0.0 & 0.0 & 0.2 & 1.5) + 2w_3(0.0 & 0.0 & 0.3 & 1.1) - \lambda &= 0 \\
 2w_2(0.0 & 0.0 & 0.5 & 8.4) + 2w_1(0.0 & 0.0 & 0.2 & 1.5) + 2w_3(0.0 & 0.0 & 0.1 & 0.4) - \lambda &= 0 \\
 2w_3(0.0 & 0.0 & 0.6 & 5.5) + 2w_1(0.0 & 0.0 & 0.3 & 1.1) + 2w_2(0.0 & 0.0 & 0.1 & 0.4) - \lambda &= 0 \\
 w_1 + w_2 + w_3 &= 1
 \end{aligned}$$

R code to solve for weights:

```

# 1. Define the Matrix A (the coefficients)

# We use byrow = TRUE to ensure the numbers
are entered row-by-row

A <- matrix(c(
  0.0052252, 0.000430, 0.000622, -1,
  0.000430, 0.001168, 0.000208, -1,
  0.000622, 0.000208, 0.001310, -1,
  1,      1,      1,      0
), nrow = 4, byrow = TRUE)

# 2. Define the constants vector B

b <- c(0, 0, 0, 1)

# 3. Solve the system Ax = b

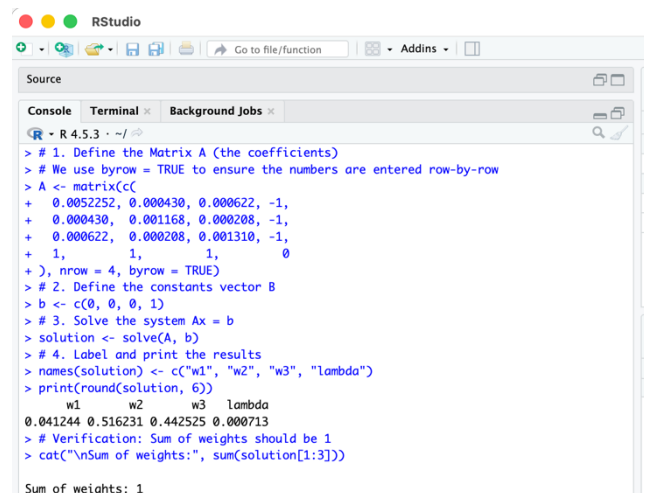
solution <- solve(A, b)

# 4. Label and print the results

names(solution) <- c("w1", "w2", "w3",
"lambda")

print(round(solution, 6))

```



```

RStudio
Source
Console Terminal Background Jobs
> # 1. Define the Matrix A (the coefficients)
> # We use byrow = TRUE to ensure the numbers are entered row-by-row
> A <- matrix(c(
+ 0.0052252, 0.000430, 0.000622, -1,
+ 0.000430, 0.001168, 0.000208, -1,
+ 0.000622, 0.000208, 0.001310, -1,
+ 1,      1,      1,      0
+ ), nrow = 4, byrow = TRUE)
> # 2. Define the constants vector B
> b <- c(0, 0, 0, 1)
> # 3. Solve the system Ax = b
> solution <- solve(A, b)
> # 4. Label and print the results
> names(solution) <- c("w1", "w2", "w3", "lambda")
> print(round(solution, 6))
      w1      w2      w3      lambda
0.041244 0.516231 0.442525 0.000713
> # Verification: Sum of weights should be 1
> cat("\nSum of weights:", sum(solution[1:3]))
Sum of weights: 1

```

Hence,

$$W_1 = W_{\text{Kenechukwu}} = 0.041244, W_2 = W_{\text{Morgan}} = 0.516231, W_3 = W_{\text{Lisa}} = 0.442525$$

B.5 Investment Alternatives for a Client

The efficient frontier shows all portfolios that offer the maximum expected return for a given level of risk. However, it does not identify a single best portfolio for every investor. The most suitable portfolio depends on the client's preferences, especially their risk tolerance, return objectives, investment horizon, and any constraints such as whether short selling is allowed.

For a risk-averse client, a portfolio close to the minimum variance portfolio would be appropriate because it offers the lowest possible volatility. For a client willing to accept more risk in exchange for higher expected return, a portfolio further up the efficient frontier may be preferable. A more aggressive investor may choose a portfolio with a higher expected return and higher standard deviation, while a conservative investor may prefer a lower-risk option.

Therefore, extra information is required before making an optimal recommendation. It is necessary to know the client's utility or risk preference, as the efficient frontier only identifies the set of efficient choices rather than the single optimal one. The diagram helps illustrate this point by showing that many efficient portfolios exist, each suitable for a different type of investor.

Conclusion

In conclusion, the analysis displays that the three stocks exhibit distinct risk-return characteristics, with K Wave Media showing high volatility and negative performance, while Meta and L'Oréal provide greater stability. The low correlations between the assets highlight the benefits of diversification, allowing investors to reduce overall portfolio risk. The efficient frontier identifies the best possible risk-return combinations; however, the optimal portfolio ultimately depends on the investor's individual preferences and constraints.